Example 7.2 The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2. Find the diameter of the cylinder.

Solution Given that, C.S.A. of the cylinder =88 sq. cm

$$2\pi rh = 88$$

$$2 \times \frac{22}{7} \times r \times 14 = 88 \qquad (h=14 \text{ cm})$$

$$2r = \frac{88 \times 7}{22 \times 14} = 2$$

Therefore, diameter $= 2$ cm

Example 7.3 A garden roller whose length is 3 m long and whose diameter is 2.8 m is rolled to level a garden. How much area will it cover in 8 revolutions?

Solution Given that, diameter $d = 2.8$ m and height $= 3$ m

radius $r = 1.4$ m

Area covered in one revolution = curved surface area of the cylinder

$= 2\pi rh$ sq. units

$$= 2 \times \frac{22}{7} \times 1.4 \times 3 = 26.4$$



Fig. 7.6

Area covered in 1 revolution = 26.4 m^2

Area covered in 8 revolutions $= 8 \times 26.4 = 211.2$

Therefore, area covered is 211.2 m^2

Thinking Corner

1. When 'h' coins each of radius 'r' units and thickness 1 unit is stacked one upon the other, what would be the solid object you get? Also find its C.S.A.
2. When the radius of a cylinder is double its height, find the relation between its C.S.A. and base area.
3. Two circular cylinders are formed by rolling two rectangular aluminum sheets each of dimensions 12 m length and 5 m breadth, one by rolling along its length and the other along its width. Find the ratio of their curved surface areas.

7.2.2 Hollow Cylinder

An object bounded by two co-axial cylinders of the same height and different radii is called a 'hollow cylinder'.

Let R and r be the outer and inner radii of the cylinder. Let h be its height.

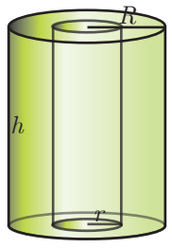


Fig. 7.7

C.S.A of the hollow cylinder = outer C.S.A. of the cylinder + inner C.S.A. of the cylinder

$= 2\pi Rh + 2\pi rh$

C.S.A of a hollow cylinder $= 2\pi(R + r)h$ sq. units

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